INTRODUCTION TO COMPUTER AIDED MANUFACTURING (CAM)



Lecture 7

Learning Objectives

The objectives of this lesson are to:

- Explain what Computer Aided Manufacturing (CAM) is
- Introduce some of the types of CAM technologies available
- Identify some benefits of CAM in manufacturing
- Explain Computer Numeric Control (CNC) machining process
- Identify some challenges with the CNC manufacturing process
- Explain 3D Printing method of manufacturing
- Study some of the 3D printing techniques available
- Look at some applications of 3D printing technology
- Identify some challenges of the 3D printing technology

Introduction

What is Computer Aided Manufacturing (CAM)?

- It is the use of computer machinery and software to automate and facilitate manufacturing processes
- It also refer to the use of a computer to **assist** in operations of a manufacturing plant such as:
  - Planning
  - Management
  - Control
  - Transportation
  - Storage



Introduction



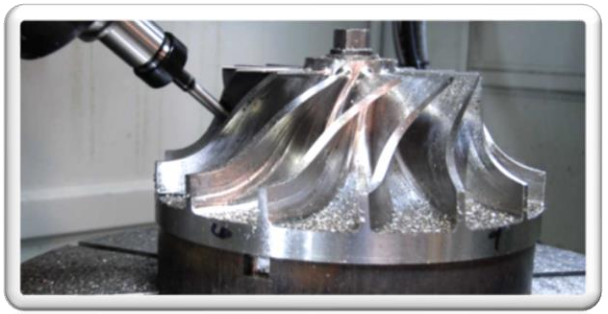
CAM Technologies

- Computer Numeric Control (CNC)
  - One most important area of CAM.
  - It is the technique of using programmed instructions to control a machine tool, which cuts, mills, grinds, punches or turns raw stock into a finished part.
- 3D Printing
  - Also known as Additive Manufacturing
  - Object created by printing successive layers

Introduction

Benefits of CAM

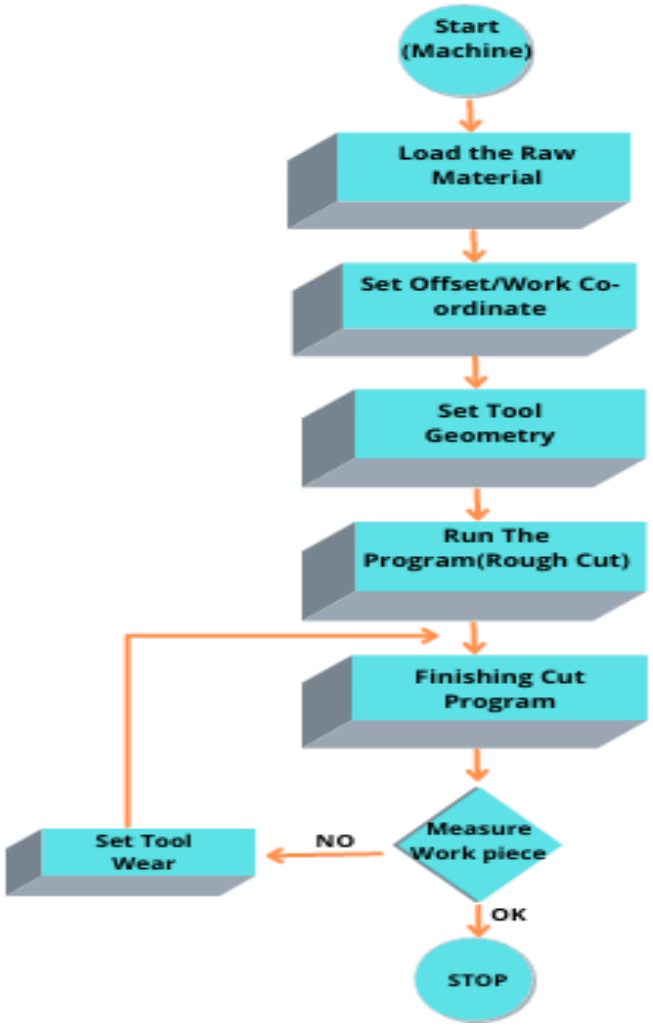
- Repetitive tasks can quickly be done
- High accuracy and precision
  - High uniformity
- Standardized production
- Increased reliability of finished products
- Manufacturing of complex shapes
- Reduce designing, evaluation and prototype wo

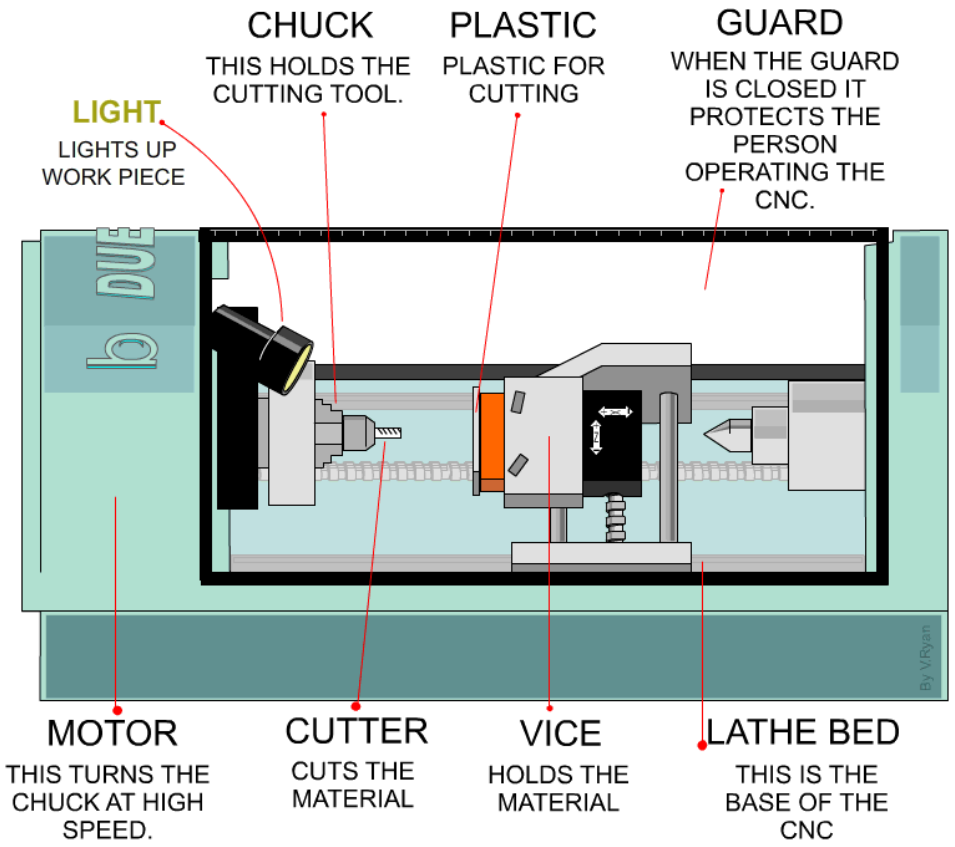




Computer Numeric Control (CNC)

Computer Numeric Control (CNC) Machining Process





Computer Numeric Control (CNC)

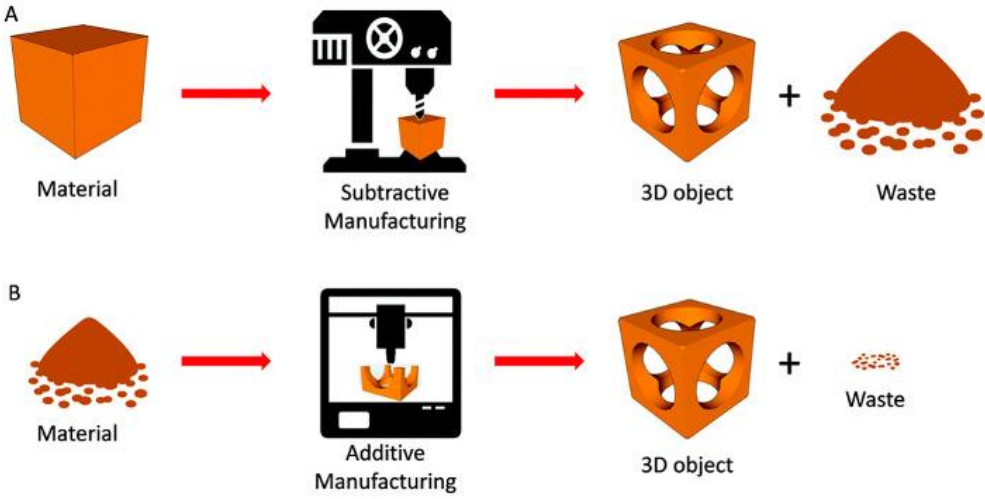
Challenges of CNC Machining Process

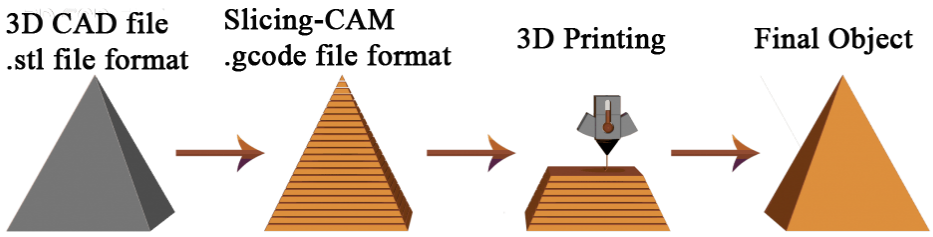
- Requires skilled personnel.
- Requires stable supply of power.
- Wrong tools or settings can result in defective product.
- Almost impossible machining some complex geometries.
- Data processing and reception from the computer system can be a bottleneck.
- High initial cost in purchasing and installing CNC machine.
- Expensive to repair.

3D Printing

What is 3D Printing?

- Additive manufacturing
- Object created by printing successive layers
- Starts with 3D objects from
  - CAD software
  - 3D scanner
  - 2D images converted to digital 3D

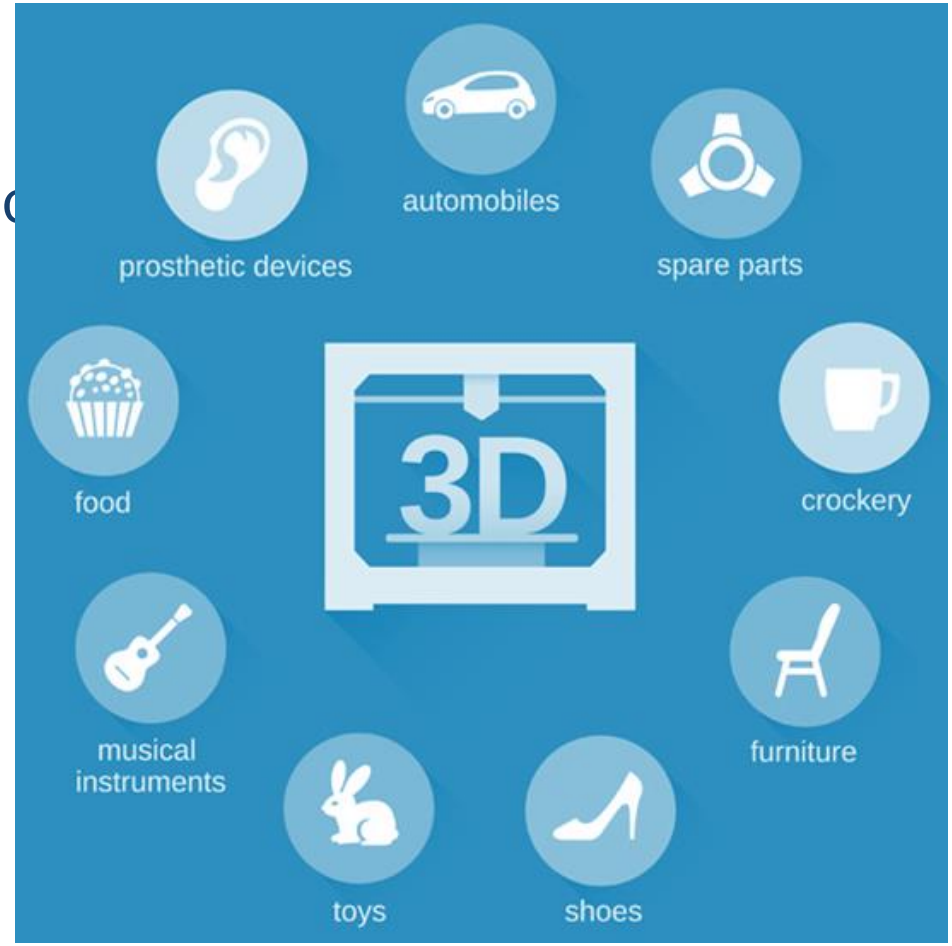




3D Printing

Why 3D Printing?

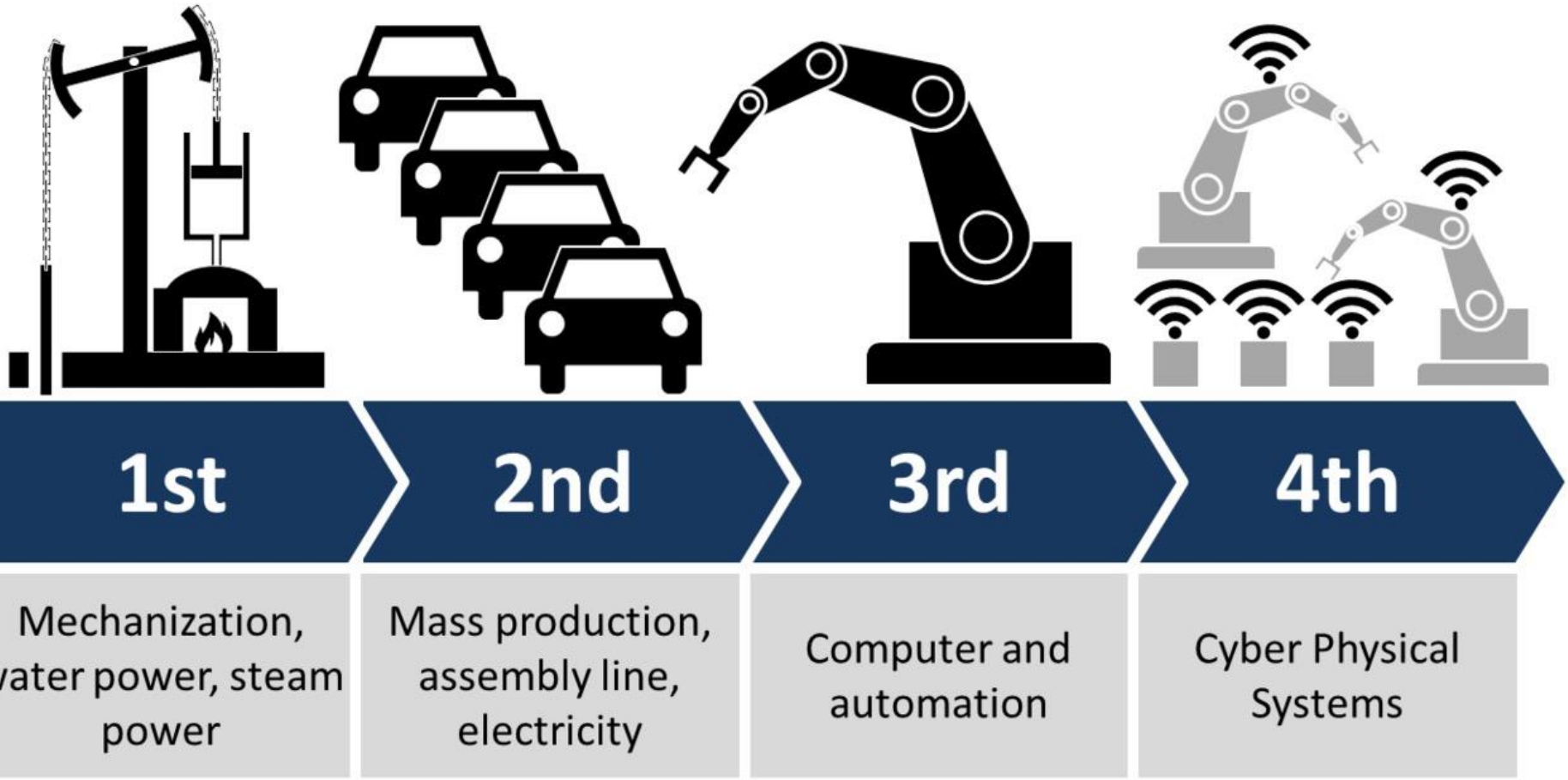
- Rapid Prototyping
- Customization and personalizatio
- Single stem manufacturing
- Design freedom
- Object complexity
- Ease of access
- Cost effective
- Localization of manufacturing



3D Printing

Why 3D Printing?

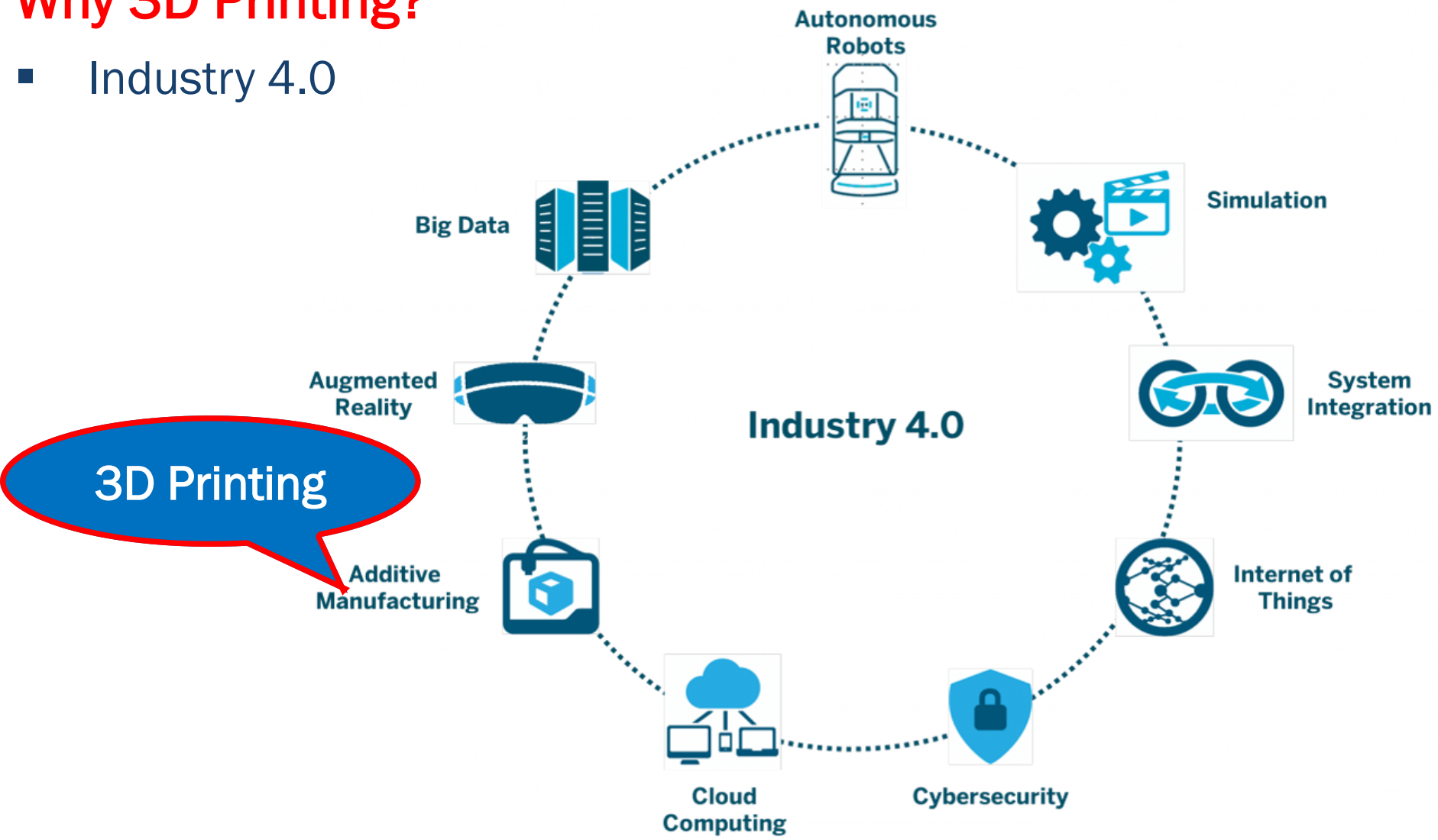
- Industrial Revolutions



3D Printing

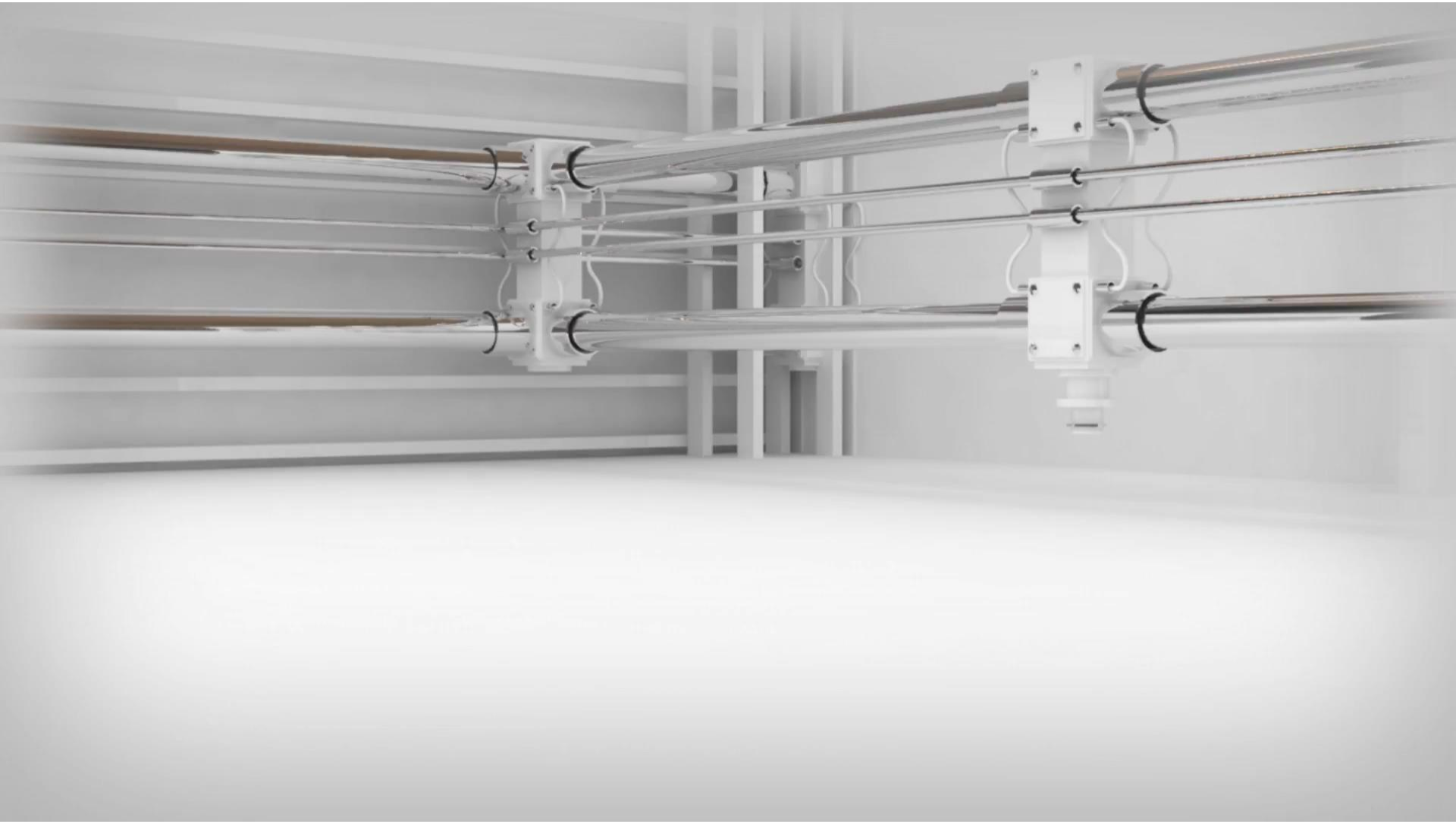
Why 3D Printing?

- Industry 4.0



3D Printing

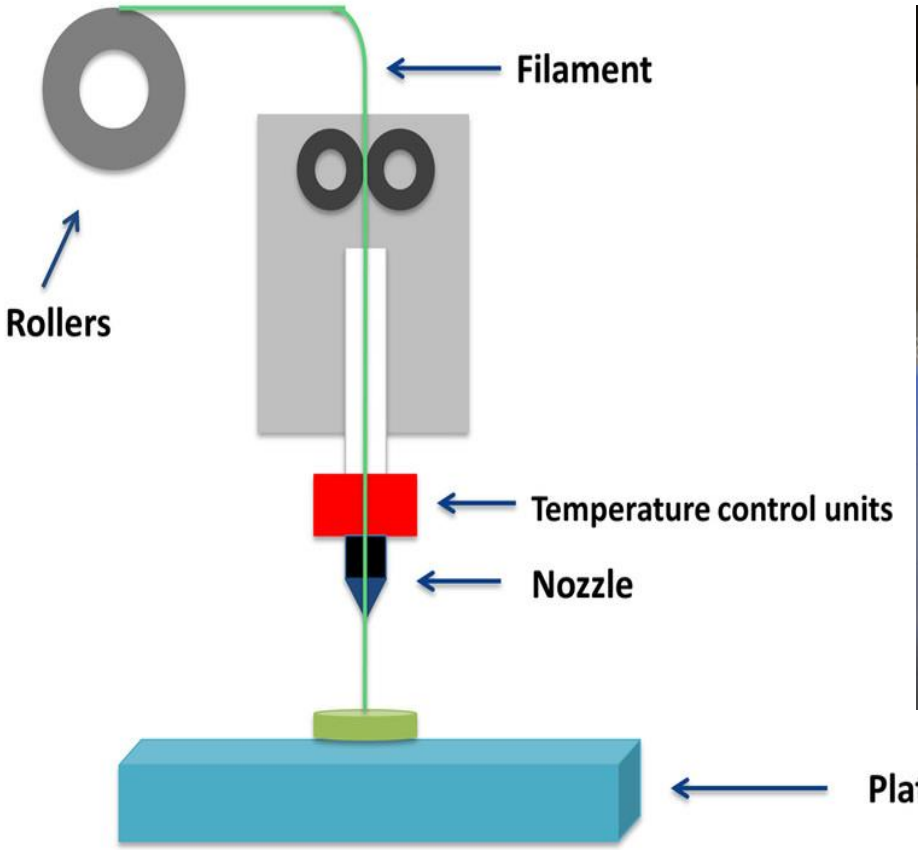
How 3D Printing Works

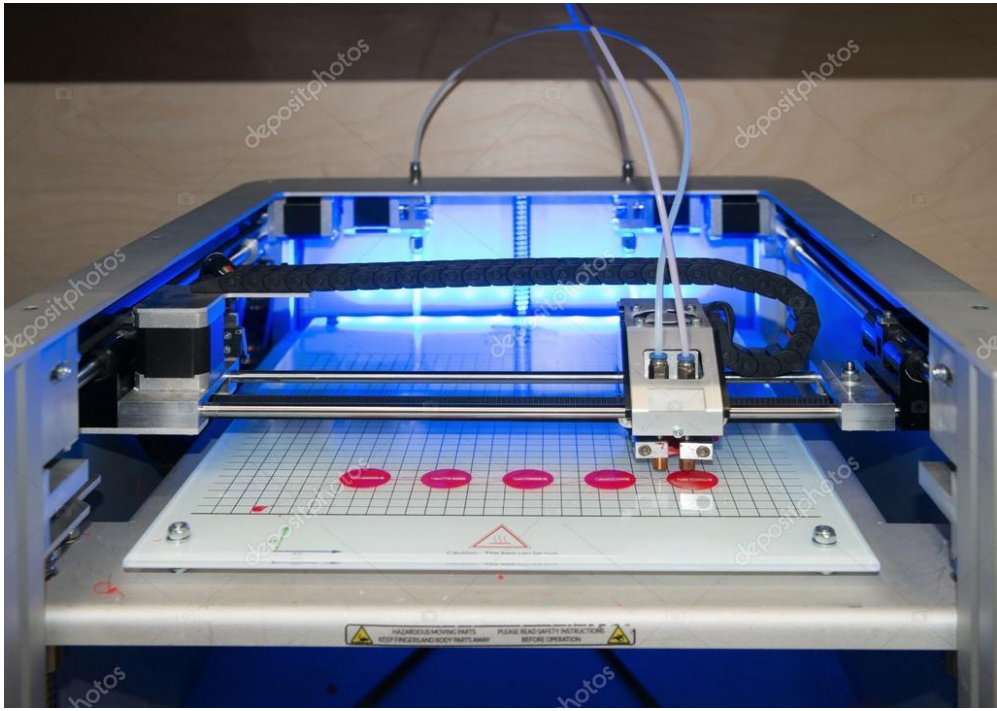


3D Printing Techniques

- There are various techniques available today

Fused Deposition Modeling (FDM)

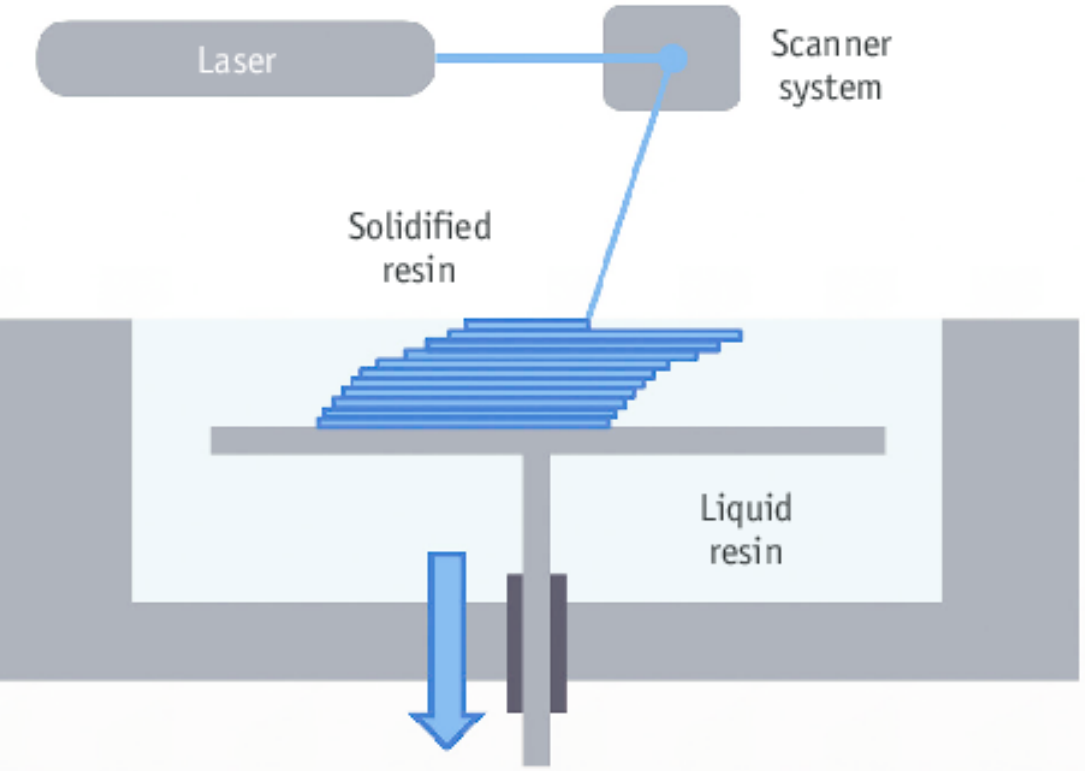


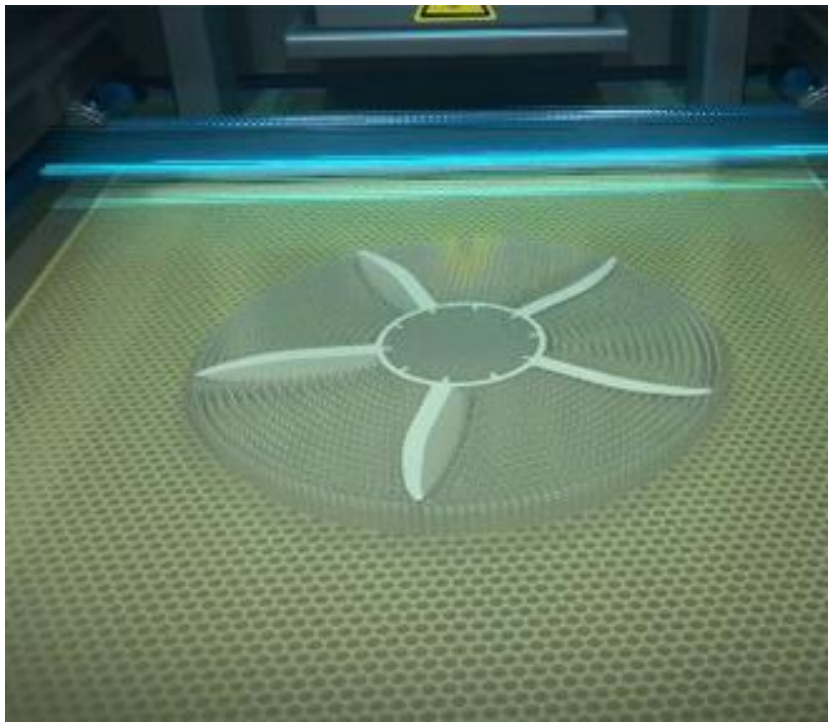


3D Printing Techniques

Stereolithography (SLA)

- Based on photopolymerisation

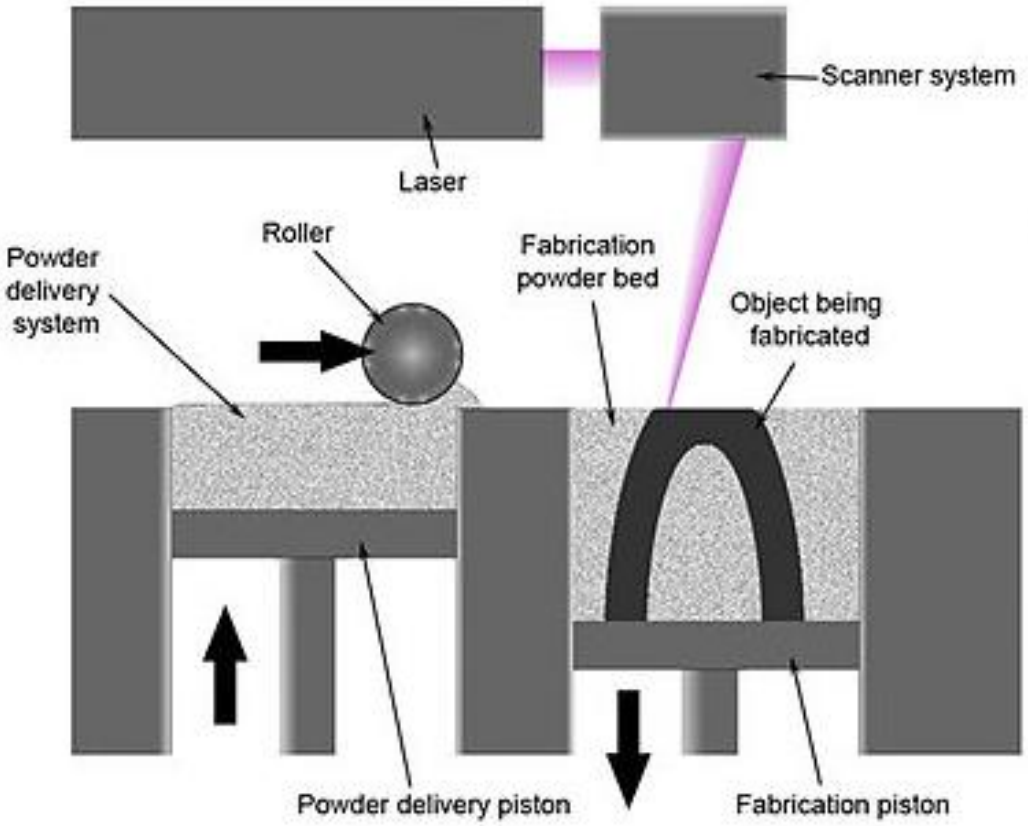




3D Printing Techniques

Selective Laser Sintering (SLS)

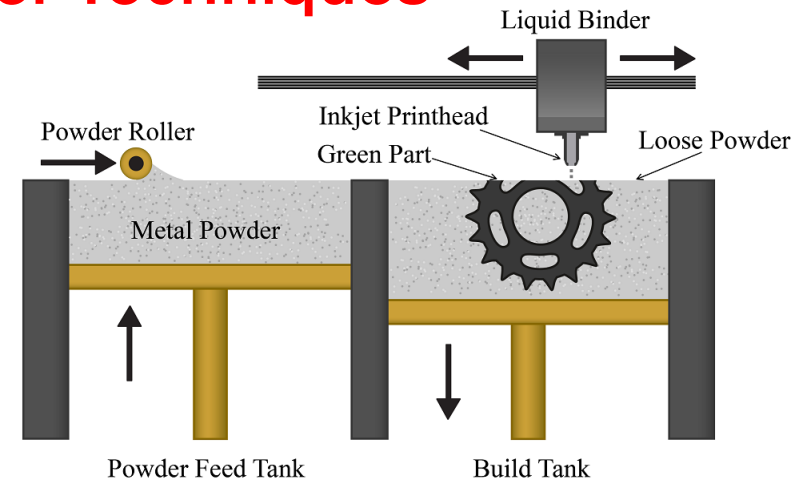
- Also known as Powder Bed Fusion (PBF)



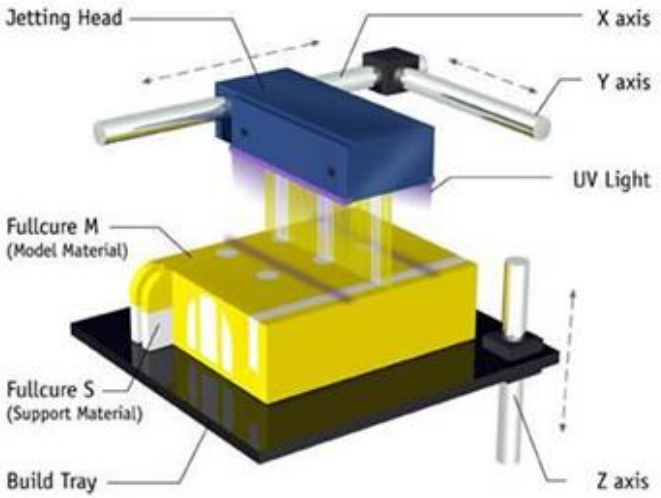


3D Printing Techniques

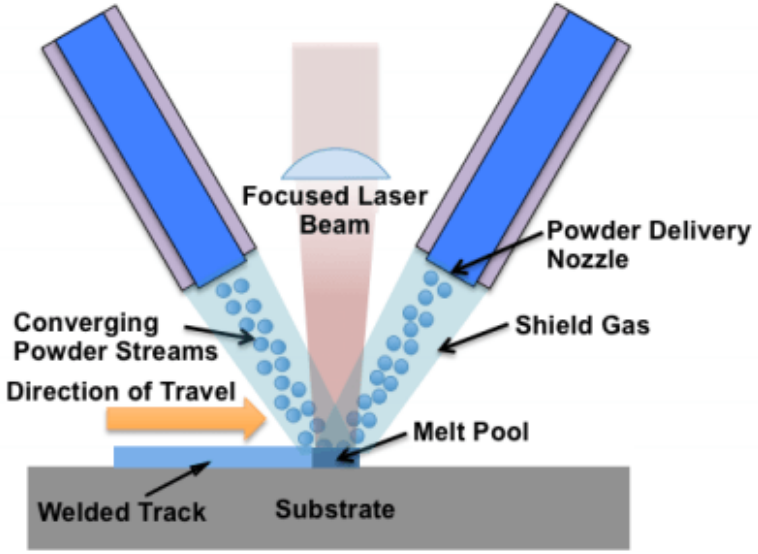
Other Techniques



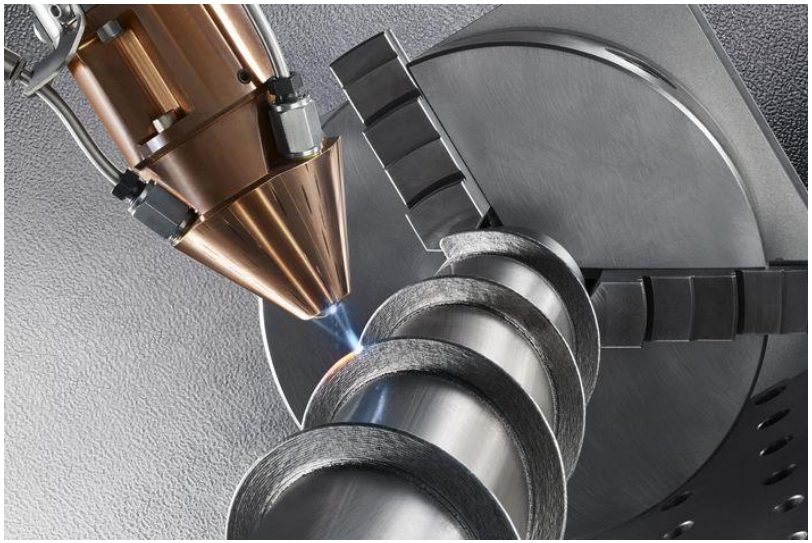
Binder jetting process



Material jetting process



Directed energy deposition (DED) process



DED 3D printing at Trumpf Company

3D Printing Applications

Manufacturing

- Cost effective where only few parts needed
- Decentralized manufacturing with Internet Of Things (IOT)
- Mass customization instead of mass production
- Rapid prototyping
- Reduction in transportation cost of parts









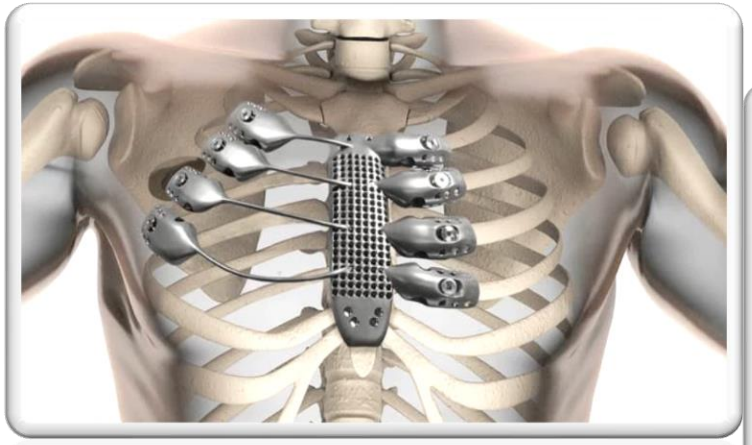
3D Printing Applications

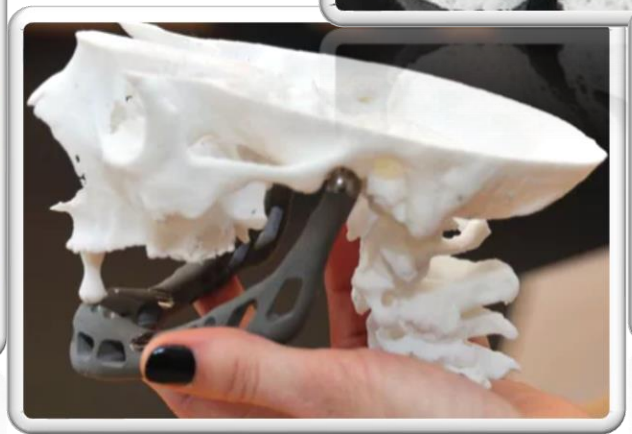
Biomedical and Medicine

- 3D printed patient-specific model for practice before surgery
- Implants and prosthetics
- Implanting of stem cells
- Single 3D printed pill according to patience's diagnostic report


heart model




Chest prosthetics





3D Printing Applications

Industrial Applications

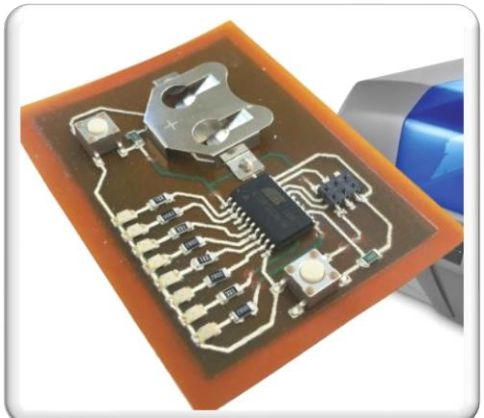
- Clothing printing
- Jewelry printing
- Smart 3D printed moulds for making jewelry
- Spare parts of vehicles and airplanes
- Architecture and Constructions
- Complex robotic shapes
- Soft sensors and actuators









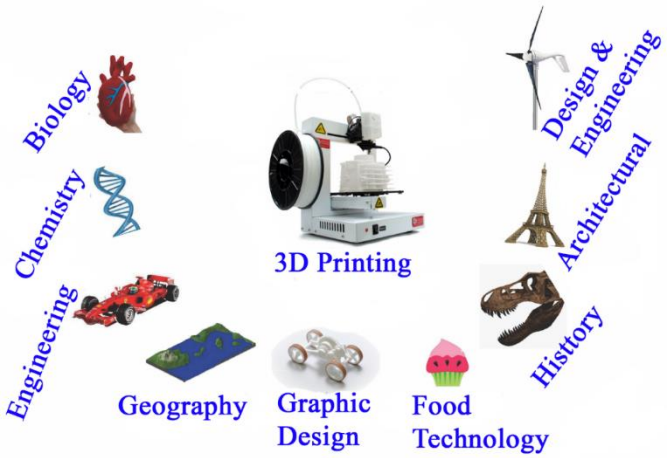




3D Printing Applications

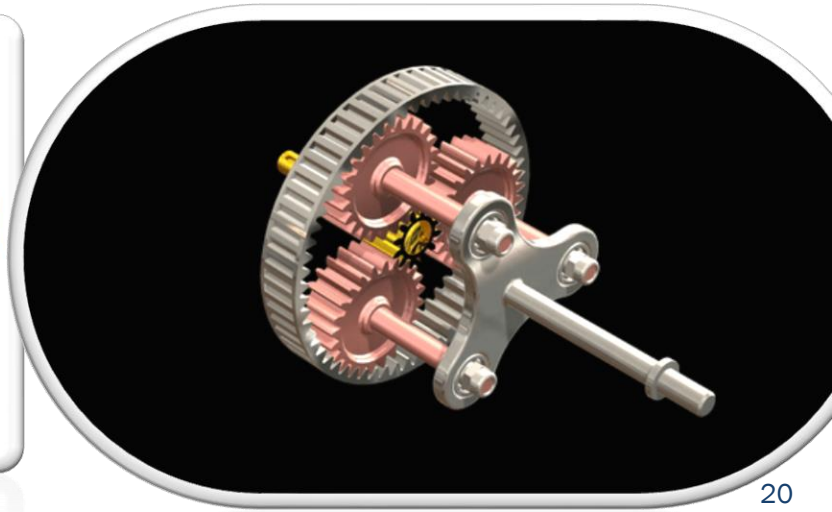
Education

- Positions students as designers and creators
- Students creates and visualize their imaginations
- Growth in special reasoning skills of students
- Helps students study with real 3D object rather than imagining









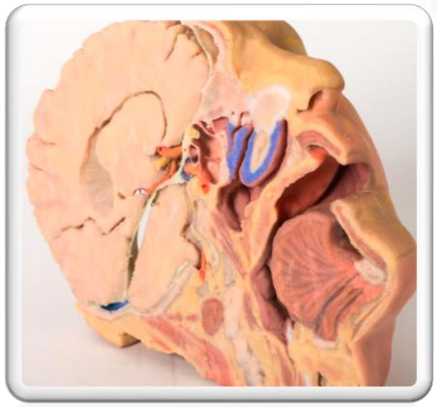
3D Printing Applications

Training

- 3D printed training resources
  - anatomical teaching resources
- Great way to improvise for unavailable and expensive devices
- Damaged parts replacement with 3D printed parts



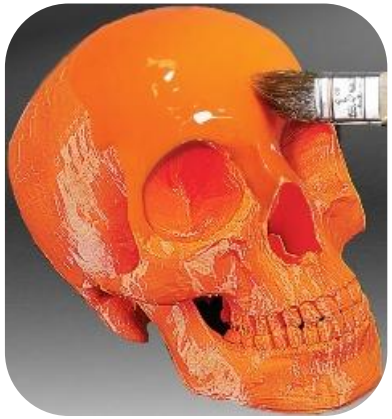


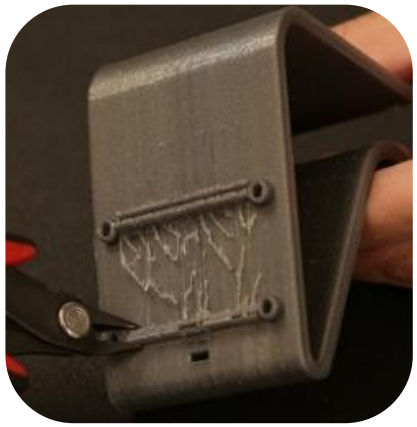




3D Printing Challenge

- Machine and material cost (**high**)
- Inconsistency and build failure (**frequent**)
- Quality assurance processes (**under-developed**)
- Build speed and volume (**low**)
- Skilled personnel (**required**)
- Cleaning & finishing (**required**)
- Intellectual property & Design theft (**high potential**)







Summary

- Computer systems and software are applied in various ways to assist in manufacturing.
- Until recently, CAM revolves mostly around CNC.
- CNC helps leverage advances in computing to aid the manufacturing sector.
- CNC follows a set of programs (usually g-code) to machine workpieces.
- 3D Printing is emerging as both a substitute and supplementary to CNC machining.
- 3D printing is able to work with a lot more complex geometries.
- 3D Printing is also promising to be cheaper and affordable than CNC machining.
- These benefits gives 3D printing a wide range of areas for its application.
- 3D printed parts however requires some post-printing treatment especially where strength is of high interest.

THANK YOU